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What is neuropsych psychoanalysis? Clinically relevant studies of the minded brain

Jaak Panksepp¹ and Mark Solms²¹ Center for the Study of Animal Well-Being, Department of Veterinary & Comparative Anatomy, Pharmacology and Physiology, College of Veterinary Medicine, Washington State University, Pullman, WA. 99164-6520, USA² Department of Psychology, University of Cape Town, Cape Town, South Africa

Neuropsych psychoanalysis seeks to understand the human mind, especially as it relates to first-person experience. It recognizes the essential role of neuroscience in such quests. However, unlike most branches of neuroscience, it positions mind and brain on an equal footing. It recognizes that the mammalian brain is not only an information processing device for behavior, but also the fount of the dynamics that is called 'mind', from joyous and sad feelings to banal cognitions and idiosyncratic flights of fancy. It is impossible to explain complex behaviors without reference to neural networks that mediate subjective mental events: that is, the causal effects of thoughts and feelings.

Neuropsych psychoanalysis accordingly counters the prevailing extreme reductionism in neuroscience and biological psychiatry. Neuromental explanatory concepts will not vanish as the brain becomes more thoroughly understood. Affective states and subjective intentionality are intrinsic to the brain. They are part of nature, exerting causal effects. Mind arises from complex brain network functions that need to be studied concurrently in humans and other animals. It is especially important to illuminate the cross-species affective foundations of the mind, given that many cognitive processes are motivated by emotional states.

Neuropsych psychoanalysis emerged during the 1990s as a response to the need to reconcile psychoanalytic and neuroscientific perspectives on the mind, with the goal of yielding a better understanding of the basic emotional foundations of psychiatric disorders, in the hope of promoting better nosology and therapeutics. Abundant research and discussions in the area have been disseminated through the house-journal of the International Neuropsych psychoanalysis Society, *Neuropsych psychoanalysis*. Since the beginning of this century, the Society has organized a dozen annual world congresses, each focusing on different heuristic themes, with prominent neuroscientists and psychoanalysts openly engaging the interfaces between neural dynamics and subjective mind. The 2012 Congress will take place in Athens, June 14–16, focusing on the topic of 'Neuropsych psychoanalytic Perspectives on Craving, Caring and Clinging'. (See <http://www.neuropsycha.org.uk> for further information.)

Neuropsych psychoanalysis is especially interested in brain functions that govern instinctual life, in particular those that are foundational for understanding subjectivity, agency and intentionality. As Freud wrote:

No knowledge would have been more valuable as a foundation for true psychological science than an approximate grasp of the common characteristics and possible distinctive features of the instincts. But in no region of psychology were we groping more in the dark. (*Beyond the Pleasure Principle*, 1920, p. 61)

However, researchers are no longer in the dark, because diverse instinctual emotional networks, which mediate rewards and punishments, have now been reliably identified across mammals [1,2]. Emerging knowledge allows one to use unconditional and conditional behavioral proxies, such as emotional vocalizations, as indicators of affective states in non-speaking animals, which in turn clarifies the elemental structure and primal sources of human and animal motivation and learning. For instance, the nature of 'reinforcement' in the brain may reflect the manner in which neural mechanisms of shifting affective states control associative learning and memory processes [3].

Neuropsych psychoanalysis, by recognizing the deep evolutionary roots of human minds and emotional disorders, seeks a more coherent understanding of the primary-process *brain* affective networks of mammalian brains than currently exists in psychological science. A key aspiration is to develop new neuropsych psychologically and neurochemically based therapies for psychiatric disorders, grounded in cross-species understandings of emotional experiences [4].

Understanding of higher cognitive functions of human brains requires strategies that also invest in neurophenomenological approaches. The human brain has an inherent capacity to generate reflexive self-reports of subjective states, and these reports provide unique and invaluable evidence of the organization of the mind (notwithstanding the almost boundless capacity of humans for self-deception and confabulation). It will never be possible to understand the reflexive tertiary processes of human brains without studying experiential verbal reports, especially in the context of pharmacological challenges and specific neuropsych psychological disorders [5,6]. Thus, neuropsych psychoanalysis offers new paradigms for the concurrent empirical analysis of brain networks, mental functions and subjective experiences in various psychiatric and neuropsych psychological disorders. It also has great interest in the relationship between spontaneous resting and/or ruminative brain states and executive controls [7,8]. Such human cognitive

Corresponding authors: Panksepp, J. (jpanksepp@vetmed.wsu.edu), (jpankse@bgnet.bgsu.edu).

studies, integrated with the cross-mammalian affective foundations of mind, can better clarify the emotional endo-phenotypes of human nature [9].

A guiding conceptualization is that the earliest affective and/or instinctual layers of mind (from homeostatic and sensory affects to primal emotions) are critical for the emergence of learning and memory (i.e. secondary processes of the brain) via so-called 'reinforcement principles'. These principles, traditionally called the 'Law of Effect', may actually be constituted by the nested hierarchies of the brain that are governed by 'Laws of Affect' (Figure 1). Thus, primary-process emotions, expressed through instinctual actions, are not unconscious. Emotional action networks in the brain generate various phenomenal states that feel good and bad in many ways, as monitored by 'rewarding' and 'punishing' learning effects. Such evolutionary 'memories' (complex unconditioned emotional responses) promote survival and reproductive success. By contrast, the secondary processes of associative learning and memory are probably deeply unconscious brain functions that parse affective states in environmental space and time, thus refining effective solutions to living. Neuropsychanalysis proposes that adaptive learning is guided by the fluctuating tides of primal neuroaffective processes. Analysis of the declarative tertiary processes in relation to these deeper layers also reveals the remarkable extent to which introspective awareness distorts the underlying causal events.

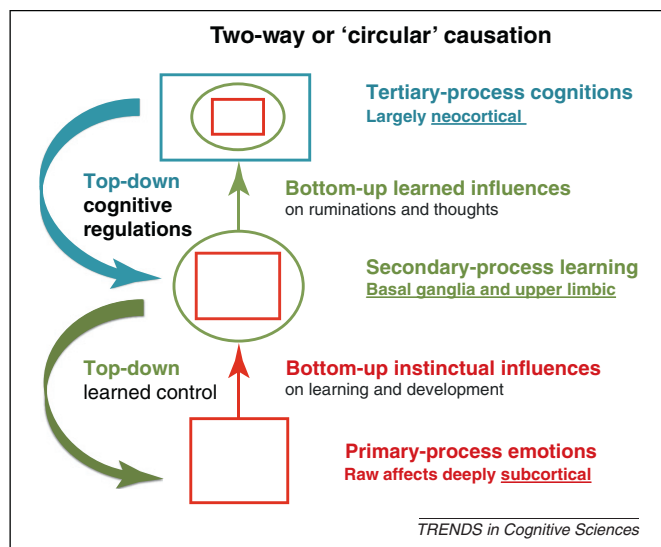


Figure 1. Nested hierarchies of circular (bottom-up to top-down) emotional control and regulation within the brain. A summary of the hierarchical two-way control is proposed to operate in every primal emotional system of the brain. The schematic suggests how higher order MindBrain functions mature and function (via bottom-up influences on the right), as they continually get integrated with the lower MindBrain functions, with primary-process affects being depicted as squares (red), secondary-process learning as circles (green) and tertiary processes, by rectangles (blue). The primary-emotional feelings are mediated by complex unconditioned response systems that control conditioning through a 'Law of Affect', which argues that the traditional concept of 'reinforcement' is a summary statement of how the neural mechanisms of affect generation control learning. The color-coding aims to convey the manner in which nested hierarchies are integrating lower brain functions into higher brain functions, which when mature, exert top-down regulatory control over behavior, yielding an overall pattern of circular causation whereby higher mind functions are coordinated by lower mind functions. Adapted from [7].

The primal affective layers of mind, most easily studied in animal models, are decisively important in the genesis of higher mental functions. Early child development largely reflects bottom-up brain-mind maturation, from primary to tertiary processes (Figure 1). For instance, early attachment problems of childhood percolate easily toward human depression [4]. By contrast, the maturing brain develops top-down controls that allow higher cognitive functions regulatory influence over lower affective ones, yielding abundant networks for the circular causality that engenders adult neuromental existence. Neuropsychanalysis explicitly recognizes the need to study closely the only species that talks about its mental experiences and cognitive perspectives. That kind of work cannot be done with other animals. With modern brain imaging, it is becoming increasingly possible to monitor simultaneously the brain correlates of these subjective reports [8].

There is much in neuropsychanalysis that is old, much that is borrowed, but also some perspectives that are new. Perhaps foremost, it seeks to understand the human mind from a cross-species evolutionary perspective, hopefully illuminating the affective roots of human nature more than traditional approaches have so far achieved. Researchers in this field assimilate the best conceptual tools and clinical observations from the pre-neuroscientific era that sought to understand the complexities of human mentation in their own right, and encourage their integrated use with all the new and old neuroscience techniques needed for a fuller understanding of mind than academic psychology and neuroscience have yet achieved. They also encourage and engage in research on the hardest problems of consciousness, from phenomenal experience (especially affective qualia) to reflexive awareness (self-referential thoughts), coordinated with preclinical studies aiming at new therapeutic practices.

So what are some of the key contributions of this fledgling approach to the mind? We note only four in this limited space: (i) the full recognition that the foundations of human emotions can be clarified in animal models, and that the neural constitution of emotional feelings can finally be empirically elucidated. Wherever in the brain investigators evoke primal emotional behaviors, those states can serve as rewards and punishments in simple learning tasks [1-3]; (ii) identification of the motivational and emotional substrates of the denial of left hemiplegias in neuropsychological patients with anosognosia. This denial is mediated by the intact left-hemisphere functions that confabulate away the concrete losses of the spatial right hemisphere, casting considerable light on the nature of self-deception [10]; (iii) clarification of the fact that dreaming and rapid eye movement (REM) sleep are dissociable, with dreaming being strongly energized by midbrain dopamine dynamics, via the general-purpose reward SEEKING urges that serve so many emotional systems [1,11,12]; and (iv) recognition of emotional feelings in animals has wide implications for understanding psychiatric disorders. For instance, neuropsychanalysis offers novel cross-species perspectives on many emotional disorders, such as addictions and depression [4,7,12-15], facilitating better preclinical models that may promote pharmacological

therapeutics more rapidly than traditional approaches [4,14,15].

With regard to the last point, preclinical models using the ‘self-report’ vocalizations of animals as indicators of their changing affective states have been developed [14]. Traditional animal models of depression are being refined, avoiding the massive external stressors that modify practically all brain functions, with more precise manipulation of particular negative affective systems that suppress the arousal of specific positive emotional systems. Such changes can be empirically monitored with robust, network-level measures of diminished positive affect [15]. These strategies can promote the development of new therapeutics, including some already in human testing [14].

In sum, neuropsychanalysis uses the best approaches of standard brain research, but does not prevaricate about the causal role of mental processes in the functions of neural networks [1–3,13]. As co-chairs of the International Neuropsychanalysis Society, we welcome all colleagues who recognize this necessity, and the power and utility of a neurophenomenal level of analysis. We trust that the neuropsychanalytic study of psychological states can profoundly enrich a fully integrated cross-species neuroscience, and thereby illuminate many mental processes in humans.

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